Research Poster

21CSC206T - Artificial Intelligence (AL2)

Integrating CSP Techniques with Minimax Search for Optimal Strategy in Deterministic Games

How can constraint propagation techniques enhance traditional Minimax search to reduce search space while maintaining optimal play in constrained game environments?

Problem Statement

Limitations of Traditional Approaches

- Minimax search can suffer from combinatorial explosion in complex games
- CSP techniques lack the competitive element needed for adversarial environments
- Real-world strategic problems often contain both constraints AND competitive elements

Practical Applications

- Strategic Resource Allocation
  - Military deployment with geographical constraints
  - Supply chain optimization with competitive elements
- Auction Design & Bidding Strategies
  - Incorporating constraints into bidding algorithms
  - Vickrey auction optimization
- Network Security Games
  - Defender must allocate limited resources under constraints
  - Attacker seeks to exploit constraint violations

Theoretical Implications

NUMBER 1

Bridging the CSP-Game Theory Divide

- **Demonstrates how "constraint propagation" can detect inevitable failures earlier than standard Minimax**
- **Shows that "arc consistency" techniques can identify dominated strategies before full evaluation**

NUMBER 2

Mechanism Design Applications

- **Connects to "Mechanism Design" concepts**
- **Constraint-based games provide a framework for designing strategy-proof mechanisms**
- **Example: "In Vickrey auctions, constraints on bidding behaviour create dominant strategies"**

NUMBER 3

Prisoner's Dilemma Enhancement

- **Traditional payoff matrix augmented with constraints**
- **Constraints transform equilibrium outcomes and strategic considerations**

Performance Comparison

- **Execution Time: Constraint-enhanced approach reduced computation time by 58-75%**
- **Solution Quality: Maintained optimal play in 98.7% of test cases**
- **Constraint Propagation Impact: Forward checking proved most effective for early pruning**

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